

AI-POWERED SMART GRID ENERGY INTELLIGENCE ASSISTANT

End-to-End System Architecture: Data Ingestion → Preprocessing → Storage → Analytics & ML → Multi-Agent System → User Delivery

1. UCI Smart Grid Dataset

2,075,259 raw 1-minute measurements (~47 mo)

- Active & Reactive Power (kW, kVAR)
- Voltage (V) & Global Intensity (A)
- Sub-metering 1 (Kitchen / Dishwasher)
- Sub-metering 2 (Laundry / Refrigeration)
- Sub-metering 3 (Water-heater & AC)

Stream / Chunk

2. Preprocessing & Feature Eng.

Validation, Cleaning & Temporal Enrichment

- Parse local wall-clock dates & times
- Handle '?' null values; drop corrupt rows
- Interpolate short gaps (<=5 min active power)
- Deduplicate records (keep latest reading)
- Temporal features: hour, day, month, iso_week,
- is_weekend, is_night, is_peak_hour (17-21)

Materialize

3. MongoDB Knowledge Base

High-Performance Indexed Collections

- energy_data: 2,075,259 cleaned rows
- hourly_energy: 34,589 hourly aggregations
- daily_energy: 1,442 daily demand records
- anomalies: 3,097 tagged anomaly events
- forecasts: Persisted forecast snapshots
- agent_traces & query_history collections

Indexed Querying

CORE ANALYTICS, ANOMALY DETECTION & FORECASTING ENGINES

Core Energy Analytics

Statistical Aggregation & Pattern Profiling

- Diurnal Hourly Demand Profiles (top hours)
- Weekday (1.036 kW) vs Weekend (1.234 kW)
- Evening Peak Surge: 18:00-21:00 (+51.4%)
- Top peak ranking (observed max: 11.122 kW)
- Monthly & seasonal variation trends

Robust Anomaly Detection

Rolling Median Absolute Deviation (MAD)

- 24-hour rolling median baseline window
- Robust Z-score scaling: (demand - med) / MAD
- Threshold: score > 3.5 flags anomalies
- Severity categorization: Low, Medium, High
- Transparent explainability & reason codes
- Controlled spike validation: Precision/Recall/F1

Time-Series Forecasting

Seasonal-Naive Baseline & Uncertainty Bands

- Captures strong 24-hour diurnal cyclicity
- Horizon support: 1 to 168 hours ahead
- Empirical prediction intervals (+/- 1.96 MAD)
- Chronological holdout validation (24-hour)
- Evaluation metrics: MAE, RMSE, MAPE
- Zero-future leakage train/test split

SPECIALIZED MULTI-AGENT ENERGY INTELLIGENCE WORKFLOW (A2A Communication & Structured Contracts)

Data Analysis Agent

Historical Patterns

- Hourly / diurnal stats
- Weekday / weekend
- Top 10 ranked peaks
- Emits structured fact msg

Anomaly Agent

Outlier Evidence

- Detects sudden spikes
- Assigns severity tags
- Calculates deviation
- Emits anomaly evidence

Forecasting Agent

Demand Predictions

- Generates N-hr forecast
- Computes lower/upper bounds
- Evaluates horizon trends
- Emits forecast array

Insight Agent

Cross-Agent Synthesis

- Aggregates evidence
- Synthesizes cross-metrics
- Computes confidence score
- Hands off to Decision

Decision Agent

Operational Support

- Peak shaving alerts
- Load shifting strategies
- Equipment inspection
- Data-backed actions

Traceability & Evidence Bundle

Evidence Grounding & Security

Zero-Hallucination Guardrails

- Prompt injection defense & input sanitization
- Strict evidence verification: numbers_are_evidence_backed
- Deterministic local fallback (works without OpenAI key)
- Multilingual intent routing (EN, ES, FR, DE, ZH, HI)
- Persisted audit trail in agent_traces collection

FastAPI Microservice

Modular REST API Backend (:8000)

- GET /api/dashboard (KPIs & coverage)
- GET /api/energy/* (trends, peaks, hourly, daily)
- GET /api/anomalies (detected events & detail)
- GET /api/forecast?hours=N (seasonal forecast)
- POST /api/query (evidence-backed NL chat)
- POST /api/agents/run & GET /api/agents/trace/{id}

JSON API

React + Vite SPA Frontend

Modern Interactive UI (:5173)

- Overview Dashboard: KPIs, active power, voltage
- Energy Analytics: diurnal curves, peak rankings
- Anomaly Detection: interactive timeline & severity
- Demand Forecast: horizon charts & error bands
- AI Assistant Chat: evidence badges & trace viewer
- Model Evaluation: MAE, RMSE, MAPE, precision/recall